

Nominal Anchors and Real Consequences: Inflation Targeting, Debt, and Investment

preliminary

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The views expressed herein are those of the authors and not necessarily those of the Federal Reserve Bank of San Francisco, or the Federal Reserve System.

secular trends in macro-finance since the 1980s often explained by real factors (demography, productivity, capital risk)

we ask: can the *monetary policy regime* itself play a role?

develop GE connections between inflation targeting and wider historical trends:

1. decline in long-term real interest rates on nominal bonds
2. increase in private (nominal) debt relative to GDP
3. investment 'drought' (declining I/Y)
4. stability of risky real returns to capital
5. strengthening of inflation targeting since the 1980s ($\phi_\pi \uparrow$)

can the monetary policy regime (5) be a unified explanation of (1)–(4) in a GE setting?

credible inflation targeting $\implies$ lower inflation volatility

- nominal bonds become *safer* in real terms
- $\implies$ inflation risk premium + interest rate risk declines $\implies R_0^* \downarrow$
- $\implies$ savers supply more savings in nominal assets $\implies \text{debt} \uparrow$
- $\implies$ capital becomes relatively *more* risky $\implies K \downarrow, \mathbb{E}_0[R_0^K] \uparrow$

successful nominal anchoring transforms the *risk characteristics* of nominal assets, triggering portfolio reallocation away from risky capital toward safer nominal bonds

monetary policy and financial instability

- Borio & White (2004), Woodford (2012), Gourio Kashyap & Sim (2018), Borio Disyatat & Rungcharoenkitkul (2019), Cairo & Sim (2023), Boissay Collard Gali & Manea (2024)

macro-finance trends

- Del Negro Giannone Giannoni Tambalotti (2017), Farhi & Gourio (2020), Eggertsson Robbins & Wold (2023)
- Jorda Schularick & Taylor (2016), Mian Straub & Sufi (2023), Laudati (2024)

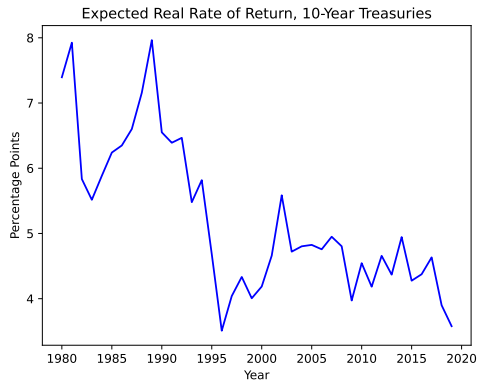
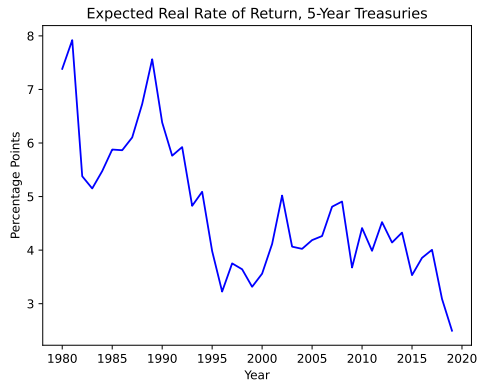
asset pricing and monetary policy

- Campbell Pflueger Viceira (2020), Miller Paron & Wachter (2023), Gourio & Ngo (2024)
[PE antecedents: e.g., Smith and Taylor 2009: ATSM under varying ϕ_π]

1. Stylized facts since the 1980s
2. A three-period macro-finance model
3. Analytical results: zero debt stochastic steady state
4. Analytical results: positive (nominal) debt
5. Conclusion

Fact 1: Decline in long-term rates

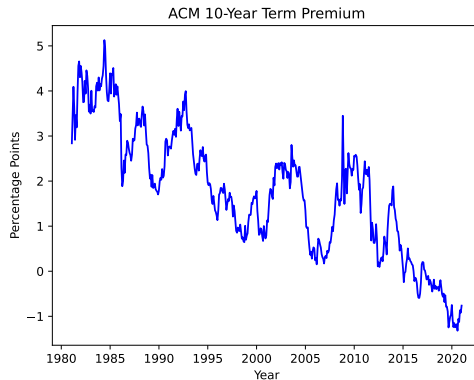
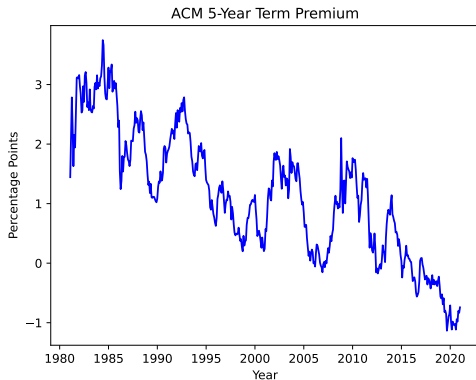
expected real yields on US nominal bonds



Expected inflation using 5-year-ahead and 10-year-ahead CPI inflation expectations from Cleveland Fed.

Fact 1: Decline in long-term rates

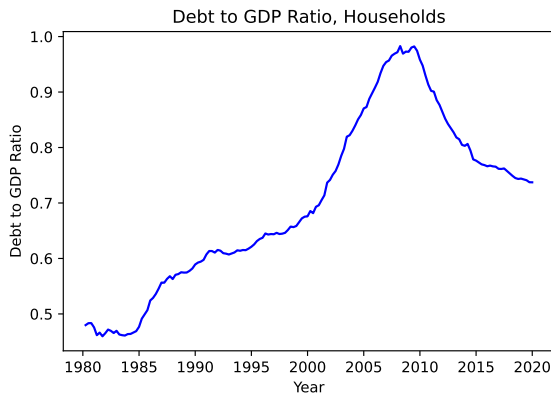
ACM term premium estimates on US nominal bonds



Term premium estimates from Adrian, Crump, and Moench (2013), Federal Reserve Bank of New York.

Fact 2: Rise in private (nominal) debt

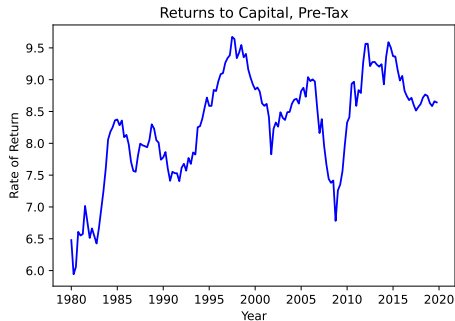
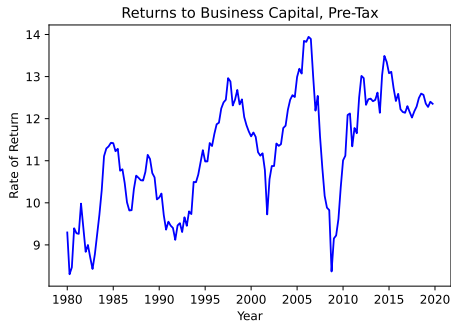
household debt relative to GDP in the US increased since 1980



Source: Our calculations based on Mian Straub & Sufi (2023)

Fact 3: Returns to capital stable or rising

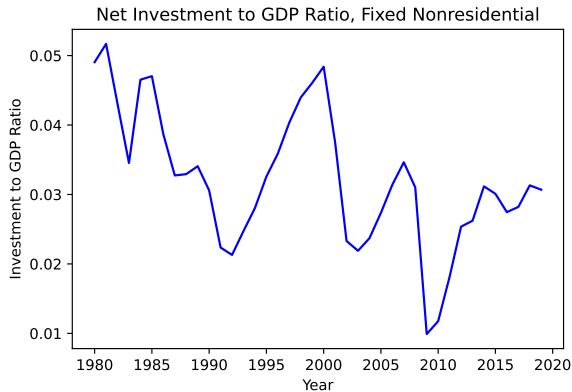
real return to capital, US data



Source: based on Gomme, Ravikumar, & Rupert (2019)

Fact 4: Investment drought

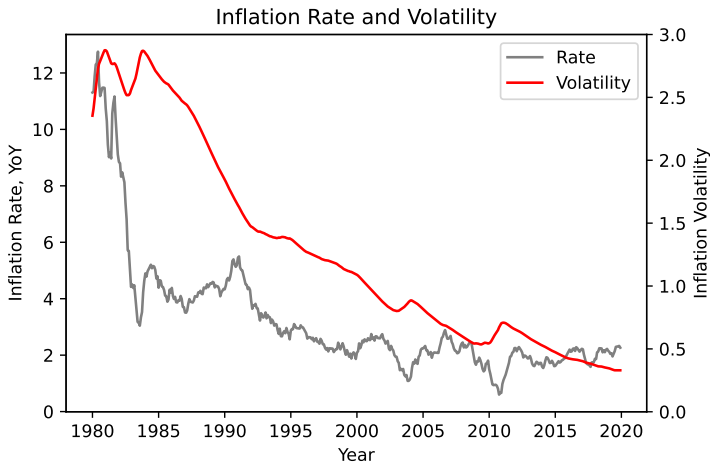
investment to GDP declined



Source: our calculations

Fact 5: Strengthening of inflation targeting

post Volcker, central banks adopt inflation targeting. US data (π , level and rolling sd)



Three Periods: ($t = 0, 1, 2$), flexible price economy

Agents

- Borrower and Savers
- Central bank (Taylor rule)

Assets

- Short-term nominal bond (zero supply)
- Long-term nominal bond
- Capital: illiquid, cannot sell at date 1

Endowments

- exogenous endowment at all dates

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Shocks

All uncertainty resolves at date 1

Aggregate Risk

- Preference shock ξ_1
- Source of inflation volatility
- inflation risk premium

Idiosyncratic Liquidity Risk on Savers

- Prob. λ : forced to liquidate at date 1
- Low income y_L in liquidity state
- interest rate risk on long-term bond

Two types $j \in \{b, s\}$: borrowers and savers, each mass one

- Borrowers: discount factor β_b^t , with β_b small (always constrained)
- Savers: discount factor $\beta^t \xi_t$, where ξ_t is a stochastic preference shock

Preference shock ξ_1 :

- $\xi_0 = 1$, $\xi_1 \sim [\xi_L, \xi_H]$ with $\mathbb{E}[\xi_1] = 1$, $\text{Var}(\xi_1) > 0$
- Date-2 persistence: $\xi_2 = \xi_1^\rho$ with $\rho \in (0, 1)$
- All uncertainty resolves at date 1

Idiosyncratic Liquidity Shock

Each saver draws $\eta \in \{0, 1\}$ (wt on date 2 utility) at beginning of date 1, i.i.d., independent of ξ_1 :

	Impatient ($\eta = 0$)	Patient ($\eta = 1$)
Probability	λ	$1 - \lambda$
Date-1 endowment	y_L	$y_H > y_L$
Long bonds at date 1	Sells at Q_1^L	Holds to maturity

Saver's date-0 utility:

$$U_s = u(C_{s,0}) + \beta \mathbb{E} \left[\xi_1 \left(\lambda u(C'_{s,1}) + (1 - \lambda) \left[u(C^P_{s,1}) + \beta \xi_1^{\rho-1} u(C^P_{s,2}) \right] \right) \right]$$

$$u(C) = C^{1-\gamma}/(1-\gamma)$$

1. **Short-term nominal bond** — issued at date 0, pays $(1 + i_0^S)$ at date 1
 - Zero supply: pins short-term real interest rate
 - Real return: $(1 + i_0^S)/\Pi_1$ — exposed to date-1 inflation
2. **Long-term nominal bond** — issued at date 0 at price Q_0^L , pays 1 at date 2
 - Liquid: can be resold at date 1 at price $Q_1^L = \frac{1}{1+i_1^S}$
 - Impatient type sells at $Q_1^L \Rightarrow$ exposed to both inflation risk and interest rate risk
3. **Real capital** K_1 — invested at date 0, produces $\bar{A}K_1^\alpha$ at date 2
 - Illiquid: cannot be sold or pledged at date 1
 - No inflation exposure

Saver's Budget Constraints

Date 0 (before learning η):

$$P_0 C_{s,0} + P_0 K_1 + Q_0^L b^L = P_0 Y_0$$

Date 1, Impatient type (λ): sells all long bonds, consumes everything

$$P_1 C_{s,1}^I = P_1 y_L + Q_1^L b^L$$

Date 1, Patient type ($1 - \lambda$): receives y_H , holds bonds to maturity, saves s_1

$$P_1 C_{s,1}^P + \frac{\lambda}{1 - \lambda} Q_1^L b^L = P_1 y_H$$

$$P_2 C_{s,2}^P = P_2 \bar{A} K_1^\alpha + P_2 Y_2 + \frac{b^L}{(1 - \lambda)}$$

Issue long-term nominal bonds, face debt limit

$$b^L/P_0 \leq d_0/R_0^{L*}$$

- Consume at dates 0 and 2 only:

$$P_0 C_{b,0} = P_0 Y_{b,0} + Q_0^L b^L, \quad P_2 C_{b,2} = P_2 Y_{b,2} - b^L$$

- Always constrained (β_b small)

Taylor rule:

$$1 + i_1^S = \frac{1}{\beta} \Pi_1^{\phi_\pi}, \quad \phi_\pi > 1$$

- policy regime characterized by $\phi_\pi > 1$
- normalize $P_0 = 1$
- Assume $\Pi_2 = \Pi_1^\delta$; $\delta \geq 0$ (Krugman 1998 special case: $\delta = 0$)

Date-1 long bond price:

$$Q_1^L = \frac{1}{1 + i_1^S} = \beta \Pi_1^{-\phi_\pi}$$

- ⇒ Links monetary policy to liquidity risk: Q_1^L fluctuates with ξ_1 through Π_1
- ⇒ More aggressive targeting stabilizes the resale price of long bonds

Three pricing equations:

- **Short bond** (no supply):

$$1 = \mathbb{E} \left[(\lambda M'_{01} + (1 - \lambda) M^P_{01}) \frac{1 + i_0^S}{\Pi_1} \right]$$

- **Long bond** (hybrid: impatient sells at date 1, patient holds to date 2):

$$Q_0^L = \mathbb{E} \left[\lambda M'_{01} \frac{Q_1^L}{P_1} + (1 - \lambda) M^P_{02} \frac{1}{P_2} \right]$$

- **Capital** (only patient types benefit, liquidity tax $(1 - \lambda)$):

$$1 = (1 - \lambda) \mathbb{E} [M^P_{02} R_K], \quad R_K = \alpha \bar{A} K_1^{\alpha-1}$$

allocations, prices, interest rates such that

1. households optimize,
2. bond markets clear, and
3. the Taylor rule holds

Zero-debt benchmark

Set $d_0 = 0$: no borrowing or lending. savers only save in capital.

from patient type's date-1 Euler + Taylor rule, solve for equilibrium inflation:

$$\Pi_1 = \bar{D} \xi_1^a, \quad \text{where } a \equiv \frac{1 - \rho}{\phi_\pi}, \quad \bar{D} \equiv \left(\frac{Y_2}{Y_H} \right)^{\gamma/\phi_\pi}$$

inflation variance:

$$\text{Var}(\log \Pi_1) = \left(\frac{1 - \rho}{\phi_\pi} \right)^2 \text{Var}(\log \xi_1) \implies \text{strictly decreasing in } \phi_\pi$$

cumulative price level: $P_2 = \Pi_1 \Pi_2 = \Pi_1^{1+\delta}$

$$\text{Var}(\log P_2) = (1 + \delta)^2 a^2 \text{Var}(\log \xi_1) \quad \text{amplified by persistence } \delta$$

Zero-debt: short-bond inflation risk premium

expected real return on the short nominal bond:

$$\bar{R}_0^{S*} = \bar{R}_1^f \cdot \underbrace{\frac{\mathbb{E}[\xi_1^{-a}]}{\mathbb{E}[\xi_1^{1-a}]}}_{\text{IRP}^S > 1}$$

- $\uparrow \phi_\pi \implies \downarrow \bar{R}_0^{S*}$ (lower compensation for inflation risk)
- as $\phi_\pi \rightarrow \infty$: $\text{IRP}^S \rightarrow 1$. Perfect IT eliminates inflation risk premium
- IRP^S independent of $\gamma, \lambda, \omega, \delta$

Decomposing the real bond yield

use bond pricing equation $1 = \mathbb{E}[M \cdot (1 + i_0^S)/\Pi_1]$:

$$R_0^{S*} = \underbrace{\frac{1}{\mathbb{E}[M_{01}]}_{R^f}} + \underbrace{\left(- \frac{\text{Cov}(M_{01}, (1 + i_0^S)/\Pi_1)}{\mathbb{E}[M_{01}]} \right)}_{\text{inflation risk premium } \psi}$$

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at $b^L = 0$: $C_{s,1}$ is deterministic for each type $\implies R^f = \bar{R}_1^f$ is invariant to ϕ_π

$$\uparrow \phi_\pi \implies \downarrow \psi \implies \downarrow R_0^{S*}$$

Why does ϕ_π lower the risk premium?

bonds as bad hedges:

- high ξ_1 = high marginal utility **and** high inflation
- $\implies$ nominal bonds pay poorly when consumption is most valuable
- $\implies$ positive inflation risk premium

but *which component* of the covariance responds to ϕ_π ?

Channel	Depends on ϕ_π ?	Contribution
Risk-free rate $\bar{R}_1^f$	No	0%
SDF volatility $\sigma_M \propto \sigma$	No	0%
Correlation $\text{Corr}(M, \Pi_1^{-1}) = -1$	No	0%
Inflation vol $\sigma_{\Pi^{-1}} \propto a \sigma$	Yes	100%

safe asset demand story: $\uparrow \phi_\pi$ reduces volatility of real bond payoffs $\implies \bar{R}_0^{S*} \downarrow$
bonds remain bad hedges (correlation = -1)

Expected real return on the long bond

$$\begin{aligned}\bar{R}_0^{L*} &\equiv \frac{\mathbb{E}[\text{real payoff}]}{\text{real cost}} = \frac{\mathbb{E}[1/P_2]}{Q_0^L/P_0} \\ &= \frac{\mathbb{E}[\Pi_1^{-(1+\delta)}]}{Q_0^L} \quad (\text{since } P_2 = \Pi_1^{1+\delta}, P_0 = 1) \\ &= \frac{\tilde{D}^{-(1+\delta)} \mathbb{E}[\xi_1^{-a(1+\delta)}]}{Q_0^L} \quad (\text{since } \Pi_1 = \tilde{D} \xi_1^a)\end{aligned}$$

Proposition: $\partial \bar{R}_0^{L*} / \partial \phi_\pi < 0$

Short-bond IRP:
$$\text{IRP}^S = \frac{\mathbb{E}[\xi_1^{-a}]}{\mathbb{E}[\xi_1^{1-a}]}$$

Long-bond IRP (patient, hold-to-maturity):
$$\text{IRP}^L = \frac{\mathbb{E}[\xi_1^{-a(1+\delta)}] \mathbb{E}[\xi_1^\rho]}{\mathbb{E}[\xi_1^{\rho-a(1+\delta)}]}$$

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Define $\Phi(\delta) \equiv \log \text{IRP}^L - \log \text{IRP}^S$:

- (i) $\Phi(0) < 0$: long bond has *smaller* IRP
- (ii) $\Phi'(\delta) > 0$: long-bond IRP strictly increasing in persistence
- (iii) $\exists! \delta^*$: for $\delta > \delta^*$, standard ordering $\text{IRP}^L > \text{IRP}^S$

Zero-debt: capital is independent of ϕ_π

capital Euler (only patient types benefit; liquidity tax $1 - \lambda$):

$$1 = (1 - \lambda) \mathbb{E} \left[M_{02}^P R_K \right], \quad R_K = \alpha \bar{A} K_1^{\alpha-1}$$

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at $b^L = 0$: $M_{02}^P = \beta^2 \xi_1^\rho (Y_2 / C_{s,0})^{-\gamma}$

- R_K is independent of ϕ_π : capital has no inflation exposure
- R_K is increasing in λ : liquidity tax requires higher compensation
- R_K is independent of ω : capital priced by patient type only

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Illiquidity premium: $R_K - \bar{R}_0^{L*} > 0$

- R_K invariant to ϕ_π , while $\bar{R}_0^{L*}$ strictly decreasing
- $\implies$ **spread widens** as inflation targeting becomes more aggressive
- R_K independent of ω , but bond price increasing in $\omega \implies$ income risk amplifies

At zero debt, more aggressive inflation targeting ($\uparrow \phi_\pi$):

1. $\bar{R}_0^{S*} \downarrow$: short-bond IRP falls (inflation volatility channel, 100%)
2. $\bar{R}_0^{L*} \downarrow$: long-bond IRP falls (same mechanism)
3. **term premium compresses** (when $\delta > \delta^*$)
4. R_K unchanged: capital return pinned by real fundamentals
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Next: what happens when we turn on debt ($b^L > 0$)?

Positive debt: perturbation around stochastic steady state

expand around the **stochastic** zero-debt equilibrium (not a deterministic SS)

- inflation is stochastic even at $b^L = 0$
- $\implies$ nonzero inflation risk premium embedded from the outset
- contrasts with standard perturbation where risk premia vanish at first order

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at $b^L > 0$: impatient type's consumption becomes **stochastic**

$$C'_{s,1} = y_L + \frac{Q_1^L}{P_1} b^L = y_L + \beta \Pi_1^{-(1+\phi_\pi)} b^L / Q_0^L$$

- $C'_{s,1}$ depends on Π_1 through liquidation payoff $\propto \Pi_1^{-(1+\phi_\pi)}$
- low base $y_L \implies$ marginal utility amplified by $\omega = (y_H/y_L)^\gamma$
- M'_{01} now covaries with inflation $\implies$ interest rate risk channel activated

Two reinforcing channels

Proposition. At $b^L > 0$, $\partial \bar{R}_0^L / \partial \phi_\pi < 0$ through two channels:

(i) Inflation risk channel (present for all λ):

- $\uparrow \phi_\pi$ reduces $\text{Var}(\log \Pi_1) \implies$ compresses IRP
- same mechanism as zero-debt benchmark

(ii) Interest rate risk channel ($\propto \lambda$, amplified by ω):

- $\uparrow \phi_\pi$ stabilizes $Q_1^L = \beta \Pi_1^{-\phi_\pi} \implies$ reduces capital-gain risk
- priced by impatient type's elevated SDF (ω times patient type's)
- vanishes at $\lambda = 0$; strengthens in λ and ω

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Return volatility:

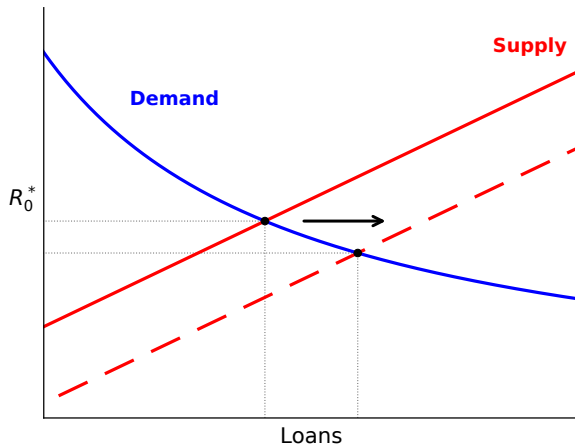
$$\frac{\text{sd}(\log R_{01}^L)}{\text{sd}(\log R_{01}^S)} = 1 + \phi_\pi$$

long bonds become *relatively* riskier as central bank is more aggressive

Intuition: supply and demand for credit

more aggressive inflation targeting $\implies$ lower R_0^{L*}

$\implies$ more nominal debt supplied by savers $\implies$ lower R_0^L , more credit in equilibrium



[binding debt limit means the demand curve is $R_0^{L*} = d_0/\text{Loans}$]

Crowd-out: the question

when bonds become safer, do savers **displace capital** or just **save more**?

write total savings as $S_0 \equiv Y_0 - C_{s,0} = K_1 + Q_0^L b^L$

differentiate with respect to bond holdings b :

$$\left. \frac{dK_1}{db} \right|_{b=0} = \underbrace{-1}_{\text{portfolio margin}} + \underbrace{\left. \frac{dS_0}{db} \right|_{b=0}}_{\text{savings margin}}$$

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- **portfolio margin** = -1 : holding S_0 fixed, each dollar of bonds displaces a dollar of capital
- **savings margin** = $1 + k$: when bonds are more attractive, saver may cut $C_{s,0}$ and save more
- $k \equiv dK_1/db$ is the **crowd-out coefficient**

Crowd-out: three forces

linearize capital Euler $(1 - \lambda) \mathbb{E}[M_{02}^P R_K] = 1$ around $b = 0$ (stochastic):

$$0 = \underbrace{(1 + k) Q_0^L \cdot \frac{\gamma}{\bar{C}_0}}_{\text{(a) date-0 smoothing}} + \underbrace{k \cdot \frac{(1 - \alpha)}{\bar{K}_1}}_{\text{(b) diminishing returns}} + \underbrace{k \cdot \frac{\gamma R_K}{\bar{C}_2}}_{\text{(c) date-2 smoothing}}$$

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(a) Date-0 consumption smoothing (drives crowd-out)

- saver saves more ($1 + k > 0$) $\implies$ cuts $C_{s,0} \implies$ higher marginal utility today
- resists further saving $\implies$ limits the savings margin

(b) Diminishing returns (absorbs displacement)

- capital falls ($k < 0$) $\implies$ marginal product R_K rises $\implies$ partially stabilizes capital

(c) Date-2 consumption smoothing (limits crowd-out)

- capital falls $\implies \bar{C}_2 = \bar{A}K_1^\alpha + Y_2$ falls $\implies$ higher marginal utility at date 2
- makes remaining capital more valuable $\implies$ resists further displacement

Crowd-out: result

Proposition. Solving for k :

$$k = -\frac{\gamma Q_0^L / \bar{C}_0}{\gamma Q_0^L / \bar{C}_0 + (1 - \alpha) / \bar{K}_1 + \gamma R_K / \bar{C}_2} \in (-1, 0)$$

strict crowd-out for all $\alpha \in (0, 1]$ and $\gamma > 0$

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Limiting cases:

- $\alpha = 1$ (constant returns): $(1-\alpha)/\bar{K}_1$ vanishes, but $\gamma R_K/\bar{C}_2$ remains
 $\implies k \in (-1, 0)$: **incomplete** even without diminishing returns
(date-2 endowment Y_2 dilutes the effect of lower capital on $\bar{C}_2$)
- $\gamma \rightarrow \infty$ (rigid consumption): savings margin $\rightarrow 0$
 $\implies k \rightarrow -1$: full displacement (saver won't adjust $C_{s,0}$)
- $\gamma \rightarrow 0$ (perfectly elastic): savings margin $\rightarrow 1$
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$\implies \gamma$ governs the **split**: high γ = more displacement, low γ = more saving

Capital extension: implications

$\uparrow \phi_\pi \implies$ safer nominal bonds $\implies b \uparrow \implies K \downarrow$ (one-for-one) $\implies \mathbb{E}_0[R_0^K] \uparrow$

mechanism: improved risk-return profile of nominal bonds crowds out capital investment

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Equilibrium implications ($\uparrow \phi_\pi$):

- $\bar{R}_0^{L*} \downarrow$ (declining real rate on nominal bonds)
- debt $\uparrow$ (rising private credit)
- $K \downarrow$ (investment drought)
- R_K stable (pinned by real fundamentals)
- $R_K - \bar{R}_0^{L*} \uparrow$ (widening safe-risky spread)
- term premium $\downarrow$ (when $\delta > \delta^*$)

CPV document: bond-stock correlation flipped sign around 2000

- they attribute this to a shift in the output-gap/inflation correlation
- aggressive MP dampens inflationary effects of supply shocks $\implies$ changes cyclical properties of bond returns

key difference: output is exogenous in our model

- we do not work through the output-gap/inflation correlation channel
- our mechanism is purely about **inflation volatility**: $\sigma_{\Pi-1} \propto 1/\phi_{\pi}$
- both papers agree: aggressive MP $\implies$ lower bond risk premia
- but through **different channels** — correlation (CPV) vs. variance (us)

our contribution: even holding the shock structure fixed, $\uparrow \phi_{\pi}$ reduces the natural rate through a **safe asset demand** mechanism

relation to existing explanations:

- our mechanism is **complementary** to real factors (demography, productivity, savings glut)
- distinct prediction: decline in real rates concentrated in **nominal assets**, while returns on **real capital** remain stable or rise

normative implications (speculative):

- IT delivers benefits: low and stable inflation
- but by making nominal bonds safer, IT encourages reallocation from capital to credit
- potential costs for long-run growth and financial stability
- optimal ϕ_π may involve trading off price stability against these considerations

Conclusion

aggressive inflation targeting regime:

- reduces inflation volatility $\implies$ lowers R_0^*
- low inflation vol & interest rate risk $\implies$ nominal bonds safer $\implies$ supply of savings in nominal assets goes up
- capital becomes relatively *more* risky $\implies$ supply of savings in real assets falls
- equilibrium: $\uparrow$ private credit, $\downarrow$ natural rate on nominal bonds, $\downarrow$ investment

a unified account of macro-finance trends since the 1980s:

declining real rates, rising debt, investment drought, stable returns on capital
all as equilibrium responses to a single shift in monetary policy regime

What next

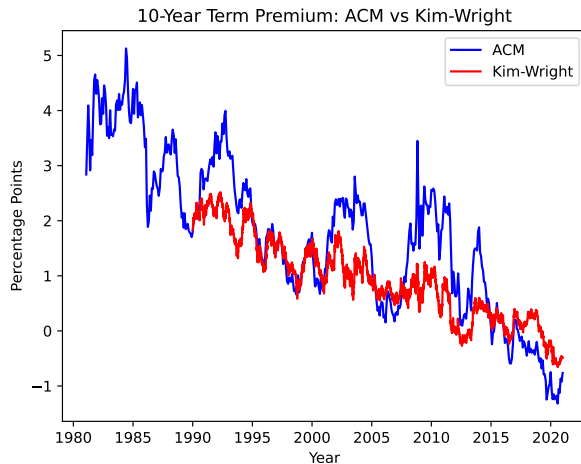
quantitative exercise:

- infinite horizon model with aggregate uncertainty
- decompose the contribution to decline in r^* from ϕ_π vs other forces (demography, productivity, capital risk)

THANK YOU!

Appendix: ACM vs Kim-Wright Term Premium

10-year term premium: ACM and Kim-Wright estimates



ACM: Adrian, Crump, and Moench (2013). Kim-Wright: Kim and Wright (2005).

Zero-debt: pricing kernels

at $b^L = 0$: date-1 consumption differs only through idiosyncratic income type-specific date-1 SDFs:

$$M'_{01} = \beta \xi_1 \left(\frac{y_L}{C_{s,0}} \right)^{-\gamma}, \quad M^P_{01} = \beta \xi_1 \left(\frac{y_H}{C_{s,0}} \right)^{-\gamma}$$

both $\propto \xi_1$ — differ only in ω :

$$M'_{01} = \omega M^P_{01}, \quad \omega \equiv \left(\frac{y_H}{y_L} \right)^{\gamma} \geq 1$$

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average sdate 1 SDF :

$$M_{01} = \lambda M'_{01} + (1 - \lambda) M^P_{01} = \frac{\xi_1}{\bar{R}_1^f}; \quad \bar{R}_1^f \equiv \frac{1}{\beta [\lambda \omega + (1 - \lambda)]} \left(\frac{y_H}{C_{s,0}} \right)^{\gamma},$$

$\implies$ **at** $b^L = 0$: average SDF is proportional to ξ_1 , regardless of λ or ω

Zero-debt: long-bond yield

long-bond Euler:

$$Q_0^L = \mathbb{E} \left[\lambda M'_{01} \frac{Q_1^L}{P_1} + (1 - \lambda) M'_{02} \frac{1}{P_2} \right]$$

at $b^L = 0$, substitute and collect powers of ξ_1 :

	SDF	$\times$	Real payoff	= ξ_1 -exponent
Impatient	$M'_{01} \propto \xi_1^1$	$\times$	$Q_1^L/P_1 = \beta \Pi_1^{-(1+\phi_\pi)} \propto \xi_1^{-a(1+\phi_\pi)}$	$\xi_1^{1-a-a\phi_\pi} = \xi_1^{\rho-a}$
Patient	$M'_{02} \propto \xi_1^\rho$	$\times$	$1/P_2 = \Pi_1^{-(1+\delta)} \propto \xi_1^{-a(1+\delta)}$	$\xi_1^{\rho-a(1+\delta)}$

where $1 - a(1 + \phi_\pi) = 1 - (1 - \rho) - a = \rho - a$

Two cases:

- $\delta = 0$: both terms involve $\mathbb{E}[\xi_1^{\rho-a}]$
- $\delta > 0$: terms involve different moments — patient type exposed to $P_2 = \Pi_1^{1+\delta}$
- $\omega > 1$ ($y_L < y_H$): impatient term has coefficient $\propto \omega$, patient term does not

Appendix: relation to alternative shock structures

baseline: $\xi_2 = \xi_1^\rho$ with $\rho < 1$ (mean reversion)

- high ξ_1 = bad times (high marginal utility) and high inflation
- bonds pay poorly in bad states $\implies$ **positive risk premium**
- $\uparrow \phi_\pi \implies \downarrow R_0^*$

alternative: $\xi_2 = \xi_1^{1+\rho}$ with $\rho > 0$ (persistence)

- high $\xi_1 \implies$ precautionary saving, deflationary pressure
- bonds are *good* hedges $\implies$ **negative risk premium**
- $\uparrow \phi_\pi \implies \uparrow R_0^*$ (opposite sign)

sign depends on whether bonds are bad hedges (baseline) or good hedges (alternative)